

KCPC Week 1 Problems

Reasoning Foundations

How To Use This Sheet

Each section has five levels. Start where you are comfortable, but try at least one problem above your comfort zone. Have a genuine attempt before looking at the solutions, and use them only when you are truly stuck. The aim is to learn from the struggle, not merely to see the answer.

1 Logic

Basic – Same Birthday Month

There are 13 students in a room. Prove that at least two of them were born in the same month.

Easy – Glove Selection

A drawer contains 5 pairs of black gloves, 3 pairs of brown gloves, and 2 pairs of grey gloves. You choose gloves in the dark.

- (a) What is the smallest number needed to guarantee at least one matching pair?
- (b) What is the smallest number needed to guarantee one matching pair of each colour?

Medium – Three People: Knights, Knaves, and Spies

Three inhabitants, Alice, Bob, and Carol, each belong to one of three types: **Knight** (always tells the truth), **Knave** (always lies), or **Spy** (may tell truth or lie). They make the following statements:

- (a) Alice says: “Bob is a Knave.”
- (b) Bob says: “Carol is a Knight.”
- (c) Carol says: “Alice is a Spy.”

Assuming exactly one is a Knight, one is a Knave, and one is a Spy, determine who is who.

Hard – Islander Statements

Ten islanders each make one statement. Islander i says: “At least i of us are liars”, for $i = 1, 2, \dots, 10$. Every islander is either a truth-teller or a liar. Which islanders are telling the truth?

Very Hard – Counterfeit Coin

You have 12 coins. Exactly one is counterfeit, and it may be heavier or lighter than the others. Using a balance scale exactly three times, describe a strategy that always identifies the counterfeit coin and whether it is heavy or light.

2 Invariants

Basic – Jigsaw Assembly

A jigsaw puzzle has 500 pieces. A move joins two existing sections into one larger section. What is the minimum number of moves needed to assemble the puzzle?

Easy – Odd Handshakes

At a party, some pairs of people shake hands. Prove that the number of people who shook hands an odd number of times is even.

Medium – Chameleons

An island has 10 brown, 14 grey, and 15 black chameleons. When two chameleons of different colours meet, both change to the third colour. Can all chameleons eventually have the same colour?

Hard – The MU Puzzle

Start with the string MI. You may repeatedly apply these rules:

- append U to any string ending in I;
- replace Mx by Mxx;
- replace III by U;
- delete UU.

Can you obtain MU?

Very Hard – The Fifteen Puzzle

The 15-puzzle has tiles numbered 1 to 15 in a 4×4 frame with one empty square. The target order has 14 and 15 in the final two positions. Is it possible to solve the same board if only tiles 14 and 15 are swapped?

3 Colouring Arguments

Basic – One Missing Corner

Can an 8×8 chessboard with one corner square removed be tiled by dominoes?

Easy – Corner-To-Corner Knight Journey

Can a knight start in the lower-left corner of a standard chessboard, visit every square exactly once, and end in the upper-right corner?

Medium – Board Walks

For each board shape below, either find a path that visits every square exactly once using horizontal/vertical moves, or prove none exists.

- (a) A cross made from a 3×9 rectangle and a 9×3 rectangle sharing their central 3×3 square.
- (b) The same cross with four squares removed: indexing rows and columns from 0 to 8 starting at the top-left corner of its 9×9 bounding box, remove $(0, 3)$, $(3, 0)$, $(3, 8)$, and $(8, 3)$.

Hard – Jumping To The Other Side

On a 5×6 board, a diagonal from the bottom-left to the top-right corner divides the cells into two sets of 15, marked A and B below. Initially each A cell has one counter and each B cell is empty.

AAAAAB
 AAAABB
 AAABBB
 AABBBB
 ABBBBB

A counter may jump over an adjacent counter to the empty cell immediately beyond it, horizontally, vertically, or diagonally; the jumped counter stays in place. Can all counters be moved onto B cells?

Very Hard – Domino Tiling Revisited

For which n can an $n \times n$ chessboard with two missing squares of opposite colours always be tiled by dominoes?

4 Induction And Monovariants

Basic – Polygon Triangulation

A convex n -gon is divided into triangles by drawing non-intersecting diagonals. Prove by induction that the number of triangles is always $n - 2$, regardless of the triangulation.

Easy – Induction Inequality

Prove that $n! > 2^n$ for every integer $n > 3$.

Medium – Infected Chessboard

A virus spreads on an $n \times n$ board. A square becomes infected if at least two of its horizontal/vertical neighbours are infected. If you may choose the initially infected squares, what is the smallest number with which the whole board can eventually become infected?

Hard – Parliament Pacification

In a parliament, each member has at most three enemies. Prove that the members can be split into two chambers so that no member has more than one enemy in their own chamber.

Very Hard – Pebble Spreading

A pebble starts at $(0, 0)$ in the non-negative quadrant of an infinite grid. A move replaces one pebble by two pebbles in the cells immediately above and immediately to the right, if both are empty. Can you clear all pebbles from the first n diagonals $i + j = 0, 1, \dots, n - 1$? Determine all possible positive integers n .

